

Topology of the real line – 1

Open and closed sets, closure and boundary, density

8 сентября 2026 года

Theoretical minimum.

Def. For $E \subseteq \mathbb{R}$ the *interior* $\text{int } E$ is the union of all open intervals contained in E , the *closure* $\overline{E}$ is the intersection of all closed sets containing E , and $\partial E = \overline{E} \setminus \text{int } E$. A set is *dense* if $\overline{E} = \mathbb{R}$ and *nowhere dense* if $\text{int } \overline{E} = \emptyset$.

Theorem 1 (structure of open sets). Every open $U \subseteq \mathbb{R}$ is an at most countable union of pairwise disjoint open intervals, namely of its connected components, and this decomposition is unique. The countability comes from the fact that disjoint intervals can be indexed by the rational points they contain. No such description exists for closed sets.

Theorem 2 (duality and the two ways to be small). $\text{int}(E^c) = (\overline{E})^c$ for every E . Consequently E is nowhere dense if and only if the complement of $\overline{E}$ is dense, if and only if every interval contains a subinterval disjoint from E .

Fact 1 (F_σ and G_δ). A countable union of closed sets is called an F_σ set, a countable intersection of open sets a G_δ set. Every open set is F_σ and every closed set is G_δ ; $\mathbb{Q}$ is F_σ and the irrationals form a G_δ .

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1. Prove Theorem 1 together with the uniqueness of the decomposition, and show that the analogue for closed sets fails: give a closed set which is not a countable union of pairwise disjoint closed intervals.
 2. (Karasev 1.47) Let $A, B \subseteq \mathbb{R}$ be non-empty and satisfy $\overline{A} \cap B = \emptyset$ and $A \cap \overline{B} = \emptyset$. Prove that there are disjoint open sets $U \supseteq A$ and $V \supseteq B$. Show by an example that the conclusion fails if only $A \cap B = \emptyset$ is assumed.

3. Prove Kuratowski's theorem: starting from one subset of $\mathbb{R}$ and applying closure and complement in any order one obtains at most 14 distinct sets. Exhibit a subset of $\mathbb{R}$ for which all 14 occur.
4. (Putnam 1990) Is $\sqrt{2}$ the limit of a sequence of numbers of the form $\sqrt[3]{n} - \sqrt[3]{m}$, where n and m run over the non-negative integers? Prove more generally that if $a_n \rightarrow \infty$ is increasing and $a_{n+1} - a_n \rightarrow 0$, then $\{a_n - a_m\}$ is dense in $\mathbb{R}$.
5. (Karasev 1.6–1.7) Prove that a bounded $f: \mathbb{R} \rightarrow \mathbb{R}$ whose graph is a closed subset of the plane is continuous, and give an example of a function with closed graph which is not continuous.